

DISTRIBUTIONALLY ROBUST EPISTEMIC RISK-OPTIMAL CONFORMAL PREDICTION

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ABSTRACT

Turning conformal prediction sets into discrete operational decisions under structured loss is central to safety-critical machine learning. Risk-Optimal Conformal Prediction (ROCP) provides a decision-theoretic answer, but it hedges out-of-set risk against an *unconstrained* global loss supremum $M(a) = \sup_{y \in \mathcal{Y}} \ell(a, y)$ and allocates coverage through a single plug-in base model, making it over-conservative and brittle under model misspecification. We propose **Distributionally Robust Epistemic Risk-Optimal Conformal Prediction (DRO-ROCP)**, which replaces the unconstrained supremum with a worst-case out-of-set risk over a covariate- and miscoverage-adaptive Kullback–Leibler (KL) ambiguity ball $\mathcal{P}_\eta(x)$ around a nominal predictive distribution. Via Donsker–Varadhan duality, the robust risk $M_{\text{DRO}}(x, a)$ admits a smooth, strictly convex one-dimensional dual, computable to machine precision in milliseconds. A dual-axis adaptive radius $\eta(x, \alpha)$ shrinks the ambiguity ball when the model is confident or the miscoverage target is large, while a loss-tail floor preserves heavy-tail safety; a complexity-gated Mondrian split-conformal step retains exact finite-sample $(1 - \alpha)$ marginal coverage under exchangeability. Across a medical-diagnosis benchmark (COVID-19 Radiography) and an autonomous-driving hazard task (BDD100K), DRO-ROCP matches or beats ROCP, RAC, and decision-agnostic conformal baselines. On BDD100K at $\alpha = 0.001$ it cuts realized loss from 4.91 to 4.37 and worst-case risk from 5.66 to 5.13; on a $10\times$ safety-critical medical loss matrix it retains 0.0% critical mistakes while the risk-averse baseline degrades to 12.95% critical mistakes and 9.36 realized loss.

1 INTRODUCTION

Deploying machine learning in safety-critical domains such as medical diagnosis and autonomous driving demands uncertainty quantification that can be turned into trustworthy downstream decisions (Angelopoulos & Bates, 2021; Bates et al., 2021; Stocker et al., 2025). Conformal prediction has become a foundational tool for distribution-free uncertainty quantification, delivering non-asymptotic coverage without restrictive parametric assumptions (Sadinle et al., 2016; Romano et al., 2019; 2020; Barber et al., 2019). By mapping a feature $x \in \mathcal{X}$ to a prediction set $C(x) \subseteq \mathcal{Y}$ that contains the label Y with probability $1 - \alpha$, it equips complex pipelines with rigorous statistical validity. Yet a prediction set alone does not tell a decision-maker how to act: deployment requires an explicit policy $\pi : 2^{\mathcal{Y}} \rightarrow \mathcal{A}$ that selects an action $a \in \mathcal{A}$ minimizing expected loss under a task-specific cost structure (Angelopoulos et al., 2022; Cresswell et al., 2024; Wang & Dobriban, 2026).

Risk-Optimal Conformal Prediction (ROCP) (Wang & Dobriban, 2026) established a decision-theoretic foundation for optimal action selection from prediction sets, extending quantile-oriented frameworks such as Risk-Averse Calibration (RAC) (Kiyani et al., 2025). ROCP minimizes a worst-case expected loss that balances the worst in-set loss $\ell_{C(x)}^{\text{in}}(a) = \sup_{y \in C(x)} \ell(a, y)$ against an out-of-set penalty $(\ell_{C(x)}^{\text{out}}(a) - \ell_{C(x)}^{\text{in}}(a))_+$ weighted by the miscoverage budget α . Despite its elegance, ROCP exhibits two failure modes in high-stakes settings:

1. **Pessimistic over-conservatism.** The out-of-set term uses an *unconstrained* global loss supremum $M(a) = \sup_{y \in \mathcal{Y}} \ell(a, y)$. When severe misclassifications are extremely costly (e.g. “No

Action” for a pathology, or a collision in driving), $M(a)$ is large and forces hedging against essentially zero-probability tail outcomes, over-inflating conservatism regardless of how confident the model is.

2. **Sensitivity to model misspecification.** Although split-conformal inference guarantees marginal coverage distribution-free, ROCP’s coverage allocation $t(x)$ and quantile targeting $\hat{Q}_t^x(a)$ rely on a single plug-in base model $f_x(y) \approx P(Y | X = x)$. When f_x is noisy or overconfident under shift (Tibshirani et al., 2019; Cauchois et al., 2020; Yang & Jin, 2026), instance-level coverage is misallocated and decision quality degrades.

To address both simultaneously, we introduce **DRO-ROCP**. It replaces the unconstrained supremum $M(a)$ and the point-estimate base model with a distributionally robust out-of-set risk evaluated over a covariate-dependent *epistemic ambiguity set* $\mathcal{P}_\eta(x) = \{q \in \Delta_Y : D_{\text{KL}}(q \parallel \hat{f}_x) \leq \eta(x, \alpha)\}$, a KL ball around a nominal ensemble distribution $\hat{f}_x$. Using Donsker–Varadhan duality we obtain a smooth one-dimensional convex dual for $M_{\text{DRO}}(x, a) = \sup_{q \in \mathcal{P}_\eta(x)} \mathbb{E}_{Y \sim q}[\ell(a, Y)]$, replacing $M(a)$ with realistic worst-case tail expectations bounded by model ambiguity. A dual-axis adaptive radius $\eta(x, \alpha) = \text{floor} + \eta_{\text{cap}}(1 - e^{-\gamma u(x)})m(\alpha)$ grows with normalized predictive entropy $u(x)$ but decays as α increases, while a loss-tail floor preserves heavy-tail safety. The robust value function is embedded into split-conformal dual-score calibration with complexity-gated Mondrian localization.

Contributions. (i) We propose DRO-ROCP, bridging statistical decision theory, distributionally robust optimization, and split-conformal inference to remove pessimistic over-conservatism while guarding against base-model misspecification. (ii) We derive a strictly convex 1D dual for $M_{\text{DRO}}(x, a)$ under KL ambiguity and embed it into inductive split-conformal calibration, preserving exact finite-sample $(1 - \alpha)$ marginal coverage under exchangeability. (iii) On COVID-19 Radiography and BDD100K, DRO-ROCP matches or outperforms ROCP, RAC, and decision-agnostic conformal baselines (LAS, APS, SOCOP); at $\alpha = 0.001$ on BDD100K it lowers realized loss $4.91 \rightarrow 4.37$ and worst-case risk $5.66 \rightarrow 5.13$, and on a $10\times$ medical penalty matrix it retains 0.0% critical mistakes where RAC reaches 12.95%. (iv) We provide a systematic ablation suite over the adaptive radius, localized calibration, out-of-set vs. in-set robustification, divergence geometry, and hyperparameters.

2 RELATED WORK

Conformal prediction for decision theory. Classical conformal prediction constructs sets $C(x)$ with marginal coverage $\mathbb{P}(Y \in C(X)) \geq 1 - \alpha$ while minimizing size (Sadinle et al., 2016; Romano et al., 2020; Barber et al., 2019). Risk-Controlling Prediction Sets (Bates et al., 2021) and Conformal Risk Control (Angelopoulos et al., 2022) extend this to bound expected losses. Connecting uncertainty to action, RAC (Kiyani et al., 2025) shows a max-min rule $\arg \min_a \ell_{C(x)}^{\text{in}}(a)$ is minimax-optimal for expected-quantile utility, and ROCP (Wang & Dobriban, 2026) studies expected loss, balancing in-set and out-of-set losses. Conformal decision-making has also been applied to human-in-the-loop settings (Cresswell et al., 2024), robust optimization (Johnstone & Cox, 2021), planning and control (Lindemann et al., 2022), off-policy and policy evaluation (Kuipers et al., 2024; Gan et al., 2025), and adaptive or anytime-valid risk control (Zecchin & Simeone, 2024; Hultberg et al., 2026). Prior decision-theoretic methods either ignore out-of-set error (RAC) or use an unconstrained supremum $M(a)$ with a point-estimate base density (ROCP); DRO-ROCP closes both gaps with a distributionally robust out-of-set term over a localized epistemic ambiguity set.

Distributionally robust optimization (DRO). DRO minimizes worst-case expected loss over an ambiguity set around a nominal distribution (Cauchois et al., 2020; Guo et al., 2024), using f -divergence, total-variation, or Wasserstein balls. Robust validation yields sets valid over f -divergence balls around the training population (Cauchois et al., 2020); conditional DRO handles label noise (Guo et al., 2024); localized online conformal prediction calibrates Bayesian optimization under misspecification (Kim et al., 2024); and multi-distribution robust conformal prediction stays valid across heterogeneous sources (Yang & Jin, 2026). These methods rarely provide finite-sample distribution-free coverage paired with set-valued predictions or a dual selector $g(x, \beta)$ for instance-wise coverage. DRO-ROCP derives a smooth 1D KL dual and embeds it into split-conformal dual-score calibration.

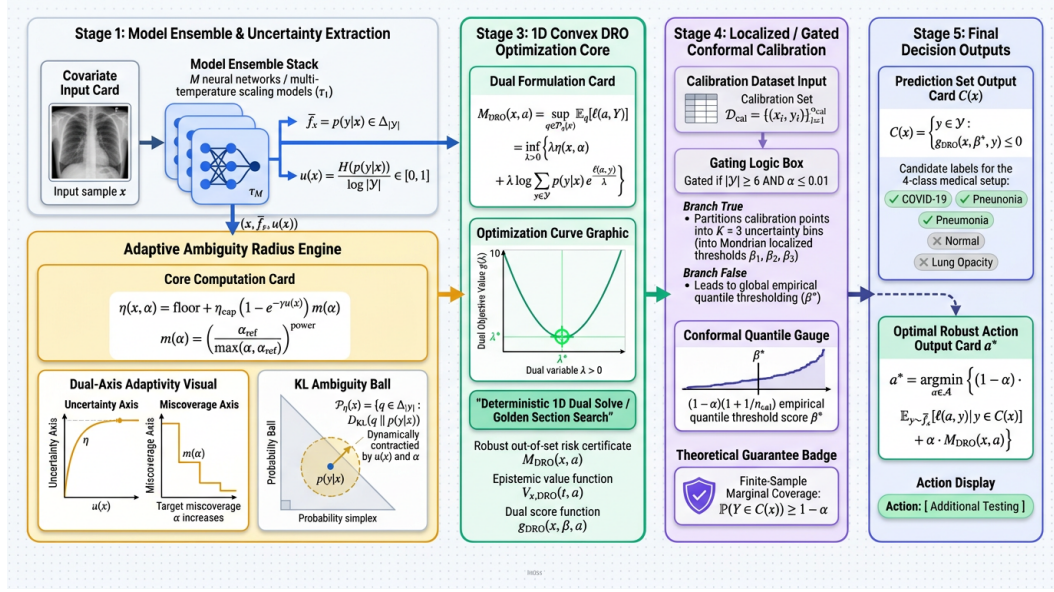


Figure 1: The DRO-ROCP framework. **Stage 1** extracts a nominal predictive distribution and normalized epistemic uncertainty from a model ensemble. **Stage 2** builds a KL ambiguity ball with a dual-axis adaptive radius driven by model uncertainty and target miscoverage. **Stage 3** solves a 1D convex dual for the robust out-of-set risk certificate and epistemic dual selector. **Stage 4** applies global or complexity-gated localized split-conformal calibration for finite-sample marginal coverage. **Stage 5** outputs the robust prediction set and the action minimizing worst-case expected loss.

Epistemic uncertainty and ensembles. Distinguishing aleatoric from epistemic uncertainty typically relies on deep ensembles (Lakshminarayanan et al., 2016; Javanmardi et al., 2025). Conformal prediction has been combined with reject options (Sokol et al., 2024) and credal sets under PAC-style guarantees (Javanmardi et al., 2025). Existing methods quantify model variance but seldom translate it into decision-theoretic action rules with finite-sample coverage. DRO-ROCP uses normalized output entropy and ensemble dispersion to scale $\eta(x, \alpha)$, directly linking epistemic uncertainty to robust decisions.

3 METHODOLOGY

3.1 PROBLEM FORMULATION AND BASELINE ROCP VULNERABILITIES

Let $\mathcal{X}$ be the feature space, $\mathcal{Y}$ a discrete label space, and $\mathcal{A}$ the action space, with a bounded non-negative loss $\ell : \mathcal{A} \times \mathcal{Y} \rightarrow \mathbb{R}_+$. A set-valued predictor maps x to $C(x) \subseteq \mathcal{Y}$. Under target miscoverage $\alpha \in (0, 1)$, ROCP (Wang & Dobriban, 2026) characterizes the worst-case expected loss over all Q with $Q(C(x)) \geq 1 - \alpha$ by the closed-form local robust loss

$$L_{C(x)}(a; \alpha) = \ell_{C(x)}^{\text{in}}(a) + \alpha(\ell_{C(x)}^{\text{out}}(a) - \ell_{C(x)}^{\text{in}}(a))_+, \quad (1)$$

where $\ell_S^{\text{in}}(a) = \sup_{y \in S} \ell(a, y)$ and $\ell_S^{\text{out}}(a) = \sup_{y \notin S} \ell(a, y)$. Optimal sets and actions follow from a pointwise value function over certified coverage $t = 1 - \alpha$,

$$V_x(t) = \min_{a \in \mathcal{A}} \{t Q_t^x(a) + (1 - t) M(a)\}, \quad (2)$$

with $Q_t^x(a) = \inf\{\theta : P_{Y \sim f_x}(\ell(a, Y) \leq \theta) \geq t\}$ the conditional loss t -quantile under f_x , and $M(a) = \sup_{y \in \mathcal{Y}} \ell(a, y)$ the unconstrained global out-of-set loss. As noted in Section 1, $M(a)$ induces *unconstrained global hedging* independent of model certainty, and reliance on a single f_x makes $Q_t^x(a)$ and $t(x)$ sensitive to misspecification.

3.2 EPISTEMIC AMBIGUITY SET AND ADAPTIVE RADIUS

DRO-ROCP centers an ambiguity set at a nominal distribution $\bar{f}_x(y) = \frac{1}{M} \sum_{m=1}^M f_x^{(m)}(y)$ from an ensemble or multi-temperature stack $\{f_x^{(m)}\}_{m=1}^M$, and measures epistemic uncertainty via normalized entropy

$$u(x) = \frac{H(\bar{f}_x)}{\log |\mathcal{Y}|} = \frac{-\sum_{y \in \mathcal{Y}} \bar{f}_x(y) \log \bar{f}_x(y)}{\log |\mathcal{Y}|} \in [0, 1]. \quad (3)$$

The KL ambiguity set is $\mathcal{P}_\eta(x) = \{q \in \Delta_{\mathcal{Y}} : D_{\text{KL}}(q \parallel \bar{f}_x) \leq \eta(x, \alpha)\}$, with a dual-axis adaptive radius

$$\eta(x, \alpha) = \text{floor} + \eta_{\text{cap}}(1 - e^{-\gamma u(x)}) m(\alpha), \quad m(\alpha) = \left(\frac{\alpha_{\text{ref}}}{\max(\alpha, \alpha_{\text{ref}})} \right)^p, \quad (4)$$

where $\gamma > 0$ controls uncertainty sensitivity, $\eta_{\text{cap}} > 0$ caps expansion, and $m(\alpha)$ (with $\alpha_{\text{ref}} = 0.02$, $p = 1$) decays the uncertainty term as α grows so large- α decisions relax toward the nominal risk. A fixed loss-tail floor $\text{floor} = \eta_0 + \kappa s(\text{tail_ratio})$ keeps protection against severe out-of-set costs even when $u(x)$ is small, preserving heavy-tail safety across all α .

3.3 DISTRIBUTIONALLY ROBUST VALUE FUNCTION AND DUAL OPTIMIZATION

Instead of $M(a)$, DRO-ROCP evaluates the out-of-set risk as a worst case over $\mathcal{P}_\eta(x)$:

$$M_{\text{DRO}}(x, a) = \sup_{q \in \mathcal{P}_\eta(x)} \mathbb{E}_{Y \sim q}[\ell(a, Y)]. \quad (5)$$

By the Donsker–Varadhan variational representation and Lagrange duality (Appendix A), this admits a smooth, strictly convex 1D dual:

$$M_{\text{DRO}}(x, a) = \inf_{\lambda > 0} \left\{ \lambda \eta(x, \alpha) + \lambda \ln \sum_{y \in \mathcal{Y}} \bar{f}_x(y) \exp\left(\frac{\ell(a, y)}{\lambda}\right) \right\}. \quad (6)$$

Strict convexity in $\lambda > 0$ lets us compute λ^* and $M_{\text{DRO}}(x, a)$ to machine precision by 1D bounded search (golden-section) in ~ 2 ms per query. Substituting yields the robust value function and dual selector

$$V_x^{\text{DRO}}(t) = \min_{a \in \mathcal{A}} \{t Q_t^{\text{DRO}}(x, a) + (1-t) M_{\text{DRO}}(x, a)\}, \quad g^{\text{DRO}}(x, \beta) = \arg \min_{t \in [0, 1]} \{V_x^{\text{DRO}}(t) - \beta t\}. \quad (7)$$

where $Q_t^{\text{DRO}}(x, a)$ defaults to the nominal quantile $\hat{Q}_t^x(a)$ (i.e. $q_{\text{scale}} = 0$; justified in Section 4.3).

3.4 SPLIT-CONFORMAL CALIBRATION AND GATED LOCALIZATION

For exact finite-sample coverage, DRO-ROCP uses inductive split-conformal calibration (Algorithm 1). On $\mathcal{D}_{\text{cal}} = \{(X_i, Y_i)\}_{i=1}^n$ it computes the dual score $E_i = \inf\{\beta \geq 0 : Y_i \in C_{\text{DRO}}(X_i, g^{\text{DRO}}(X_i, \beta))\}$, where $C_{\text{DRO}}(x, t) = \{y : \ell(a^{\text{DRO}}(x, t), y) \leq Q_t^{\text{DRO}}(x, a^{\text{DRO}}(x, t))\}$, and calibrates

$$\hat{\beta}_{1-\alpha} = \text{Quantile}\left(\{E_1, \dots, E_n\}, \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right). \quad (8)$$

Complexity-gated Mondrian calibration. When the decision space is complex ($|\mathcal{Y}| \geq 6$, e.g. BDD100K hazard states) and the target is small ($\alpha \leq 0.01$), global calibration can imbalance conditional coverage across uncertainty regimes. In this gated regime DRO-ROCP partitions $\mathcal{D}_{\text{cal}}$ into $K = 3$ entropy bins on $u(x)$ and calibrates per-bin thresholds $\hat{\beta}_{1-\alpha}^{(k)}$; otherwise it reverts to global calibration to avoid small-sample quantile variance. Marginal coverage holds in both cases (Appendix A).

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets. (i) *COVID-19 Radiography* (Kiyani et al., 2025): 4-class chest X-ray diagnosis ($\mathcal{Y} = \{\text{Normal, Pneumonia, COVID-19, Lung Opacity}\}$; $\mathcal{A} = \{\text{No Action, Antibiotics, Quarantine, Additional Testing}\}$), evaluated under the baseline loss matrix Λ_0 and a safety-critical $10\times$ variant

Algorithm 1 Distributionally Robust Epistemic Risk-Optimal Conformal Prediction (DRO-ROCP)

Require: Calibration set $\mathcal{D}_{\text{cal}} = \{(X_i, Y_i)\}_{i=1}^n$; test covariate X_{test} ; target α ; loss $\ell(a, y)$; ensemble stack $\{f^{(m)}\}_{m=1}^M$.

- 1: **for** $i = 1, \dots, n$ **do**
- 2: Extract nominal $\bar{f}_{X_i}$ and normalized entropy $u(X_i)$ (Eq. 3); set radius $\eta(X_i, \alpha)$ (Eq. 4).
- 3: Solve the 1D convex dual $M_{\text{DRO}}(X_i, a)$ for all $a \in \mathcal{A}$ (Eq. 6).
- 4: Build $V_{X_i}^{\text{DRO}}(t)$ (Eq. 7) and compute the dual score E_i (Eq. 8).
- 5: **end for**
- 6: **if** $|\mathcal{Y}| \geq 6$ **and** $\alpha \leq 0.01$ **then** partition $\mathcal{D}_{\text{cal}}$ into $K = 3$ entropy bins; compute per-bin $\hat{\beta}_{1-\alpha}^{(k)}$.
- 7: **else** compute the global quantile $\hat{\beta}_{1-\alpha}$ (Eq. 8).
- 8: Evaluate $t^* = g^{\text{DRO}}(X_{\text{test}}, \hat{\beta}_{1-\alpha})$; retrieve action $a^*(X_{\text{test}})$ and set $C(X_{\text{test}})$.

Ensure: Prediction set $C(X_{\text{test}})$ and decision action $a^*(X_{\text{test}})$.

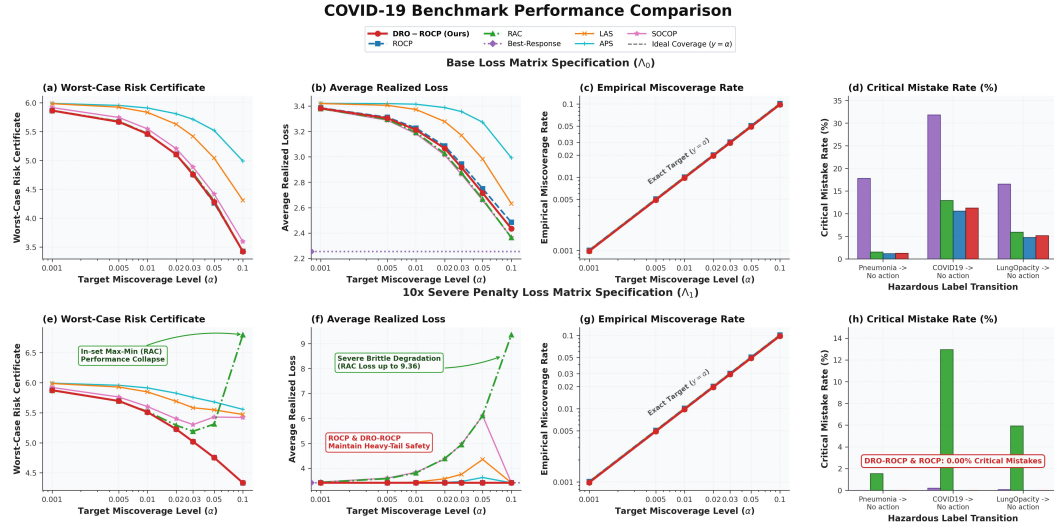


Figure 2: COVID-19 medical diagnosis under the base loss matrix Λ_0 (top) and the safety-critical $10\times$ penalty matrix Λ_1 (bottom), across target miscoverage α : worst-case risk certificate (a, e), realized loss (b, f), empirical miscoverage (c, g), and critical mistake rate at $\alpha = 0.05$ (d, h). All set methods track the target-miscoverage diagonal (c, g). Under Λ_1 , in-set risk-averse calibration (RAC) degrades sharply as α grows (e, f), whereas DRO-ROCP and ROCP retain heavy-tail safety with stable risk and zero critical mistakes (h). Mean over 20 splits; bars are ± 1 SE.

Λ_1 where severe mismatches (e.g. “No Action” for a pathology) cost $10\times$ more. (ii) *BDD100K*: a static autonomous-driving hazard task with 8 hazard states $Y \in \{0, 1\}^3$ (ahead-, left-, right-close obstacles).

Models. COVID-19 uses an ImageNet-pretrained Inception-v3 fine-tuned over 20 splits (70/10/20%); BDD100K extracts hazard probabilities from a pretrained YOLO11 detector over 20 splits (20/40/40%).

Baselines and metrics. We compare against RAC (Kiyani et al., 2025), ROCP (Wang & Dobriban, 2026), decision-calibration + best-response, and decision-agnostic sets (LAS (Sadinle et al., 2016), APS (Romano et al., 2020), SOCP) paired with the a_{ROCP} and a_{RAC} rules. We report (a) average worst-case risk certificate $\frac{1}{n} \sum_i L_{C(X_i)}(a_i; \alpha)$, (b) average realized loss $\frac{1}{n} \sum_i \ell(a_i, y_i)$, (c) empirical miscoverage $\frac{1}{n} \sum_i \mathbb{I}(y_i \notin C(X_i))$, and (d) critical mistake rates, over $\alpha \in \{0.001, 0.005, 0.01, 0.02, 0.03, 0.05, 0.1\}$.

Table 1: Main comparison on COVID-19 (Λ_0 and Λ_1) and BDD100K across $\alpha \in \{0.001, 0.01, 0.05\}$ (mean over 20 splits; SE in Appendix D). Lower is better; best per column in **bold**. Our method is shaded.

Method	COVID-19 Base (Λ_0)			COVID-19 10 $\times$ Penalty (Λ_1)			BDD100K Driving		
	$\alpha=.001$	$\alpha=.01$	$\alpha=.05$	$\alpha=.001$	$\alpha=.01$	$\alpha=.05$	$\alpha=.001$	$\alpha=.01$	$\alpha=.05$
<i>Average realized loss ($\downarrow$)</i>									
DRO-ROCP (Ours)	3.383	3.213	2.715	3.421	3.420	3.420	4.370	3.263	2.313
ROCP	3.386	3.227	2.750	3.421	3.421	3.421	4.906	3.492	2.320
RAC	3.379	3.192	2.667	3.439	3.816	6.113	4.955	4.151	2.981
LAS (a_{ROCP})	3.420	3.373	2.983	3.420	3.448	4.357	4.980	4.980	2.370
APS (a_{ROCP})	3.421	3.415	3.273	3.421	3.418	3.632	4.980	4.980	2.360
SOCOP (a_{ROCP})	3.378	3.182	2.658	3.448	3.842	6.106	4.980	4.724	3.758
Best-Response	2.255 [†]	2.255 [†]	2.255 [†]	3.423	3.423	3.423	2.308 [†]	2.308 [†]	2.308 [†]
<i>Realized worst-case risk certificate ($\downarrow$)</i>									
DRO-ROCP (Ours)	5.866	5.461	4.281	5.874	5.510	4.754	5.126	3.671	3.111
ROCP	5.863	5.460	4.263	5.874	5.509	4.752	5.664	3.837	3.136
RAC	5.869	5.470	4.303	5.870	5.520	5.310	5.728	4.683	3.126
<i>Empirical miscoverage (target = α)</i>									
DRO-ROCP (Ours)	0.0014	0.0103	0.0500	0.0015	0.0101	0.0496	0.0020	0.0094	0.0478
ROCP	0.0014	0.0104	0.0506	0.0015	0.0101	0.0497	0.0012	0.0099	0.0476

[†]Best-Response ignores set-valued uncertainty: low average loss but high critical-mistake rates (Fig. 2d, 3d).

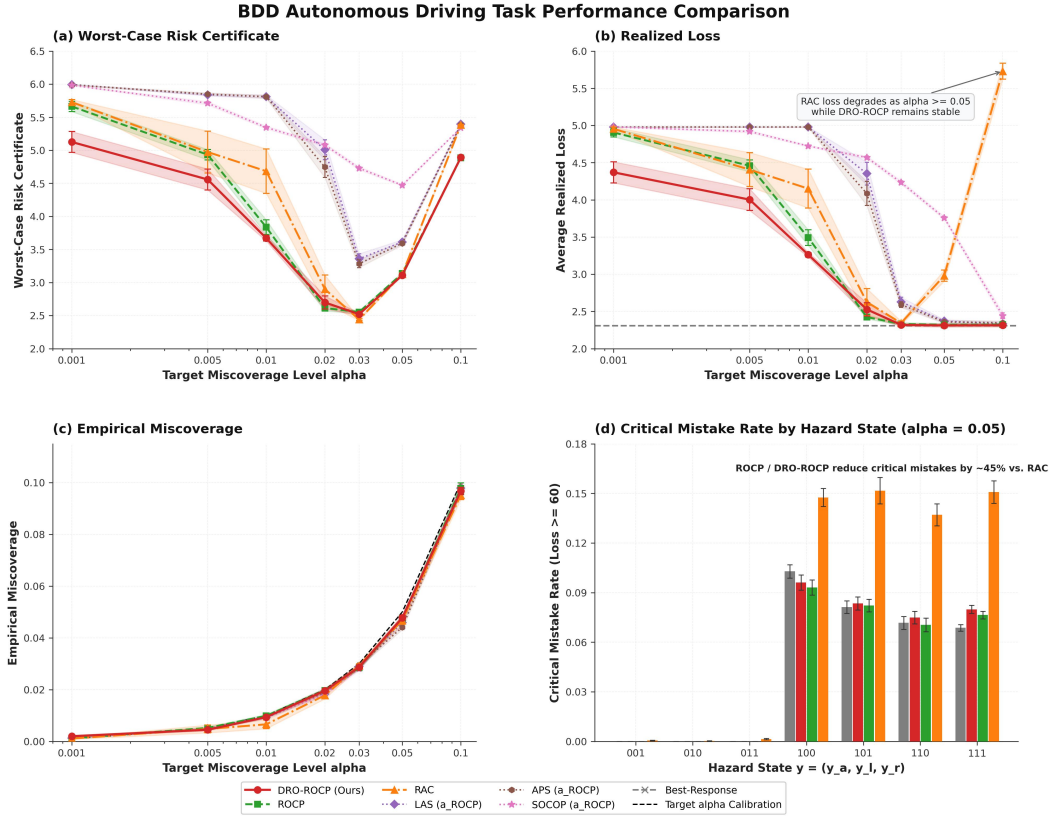


Figure 3: BDD100K driving task across α : (a) worst-case risk certificate, (b) realized loss, (c) empirical miscoverage, and (d) critical mistake rate (collision loss ≥ 60) per 3-bit hazard state $y = (y_a, y_\ell, y_r)$ at $\alpha = 0.05$. DRO-ROCP achieves the lowest risk and loss at small α and stays stable while RAC degrades for $\alpha \geq 0.05$; DRO-ROCP/ROCP cut critical mistakes by $\sim 45\%$ vs. RAC on severe states. Mean over 20 splits; bars are ± 1 SE.

4.2 MAIN RESULTS

COVID-19 base matrix. On the benign Λ_0 (Table 1, Fig. 2a–d), DRO-ROCP beats ROCP on realized loss at every α ($\alpha = 0.05$: 2.715 vs. 2.750; $\alpha = 0.10$: 2.435 vs. 2.484), matches its worst-

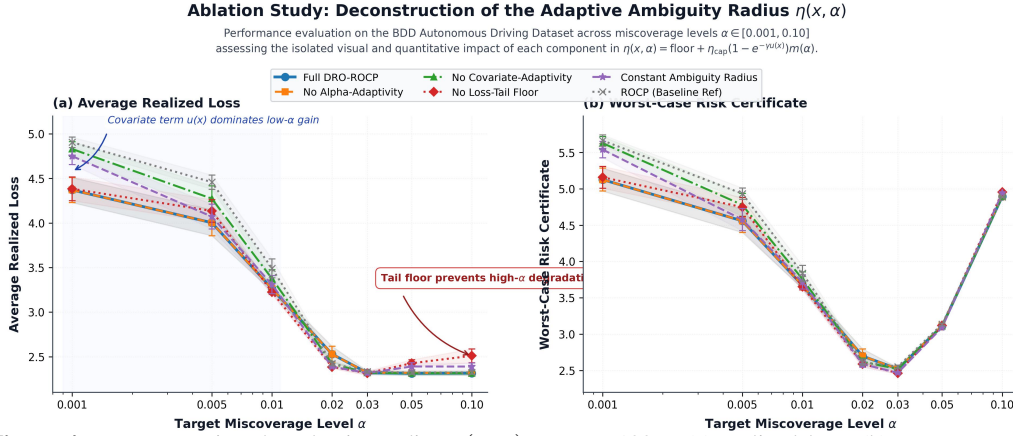


Figure 4: Deconstructing the adaptive radius $\eta(x, \alpha)$ on BDD100K: (a) realized loss, (b) worst-case risk certificate. The covariate term $u(x)$ drives the low- α gains, while the loss-tail floor prevents high- α degradation. Mean over 20 splits; bands are ± 1 SE.

Table 2: Ablation 3: out-of-set (M_{DRO}) vs. in-set (Q_t^{DRO}) robustification on COVID-19 at $\alpha = 0.05$ (20 seeds). Realized loss (RL, $\downarrow$) and COVID-19 $\rightarrow$ No-Action failure rate. Our shipped variant is shaded.

Variant	Out-of-set	In-set	Base (Λ_0)		$10\times$ (Λ_1)	
	$M(a)$	$Q_t(a)$	RL	Fail(%)	RL	Fail(%)
Full DRO-ROCP	M_{DRO}	Q_t^{DRO}	2.790	9.45	3.422	0.00
M_{DRO} only (shipped)	M_{DRO}	nominal	2.715	11.28	3.420	0.00
Q_t^{DRO} only	$\sup \ell$	Q_t^{DRO}	3.004	5.99	3.422	0.00
Standard ROCP	$\sup \ell$	nominal	2.750	10.59	3.420	0.00

case risk within one SE, and hits the coverage target (0.0500 ± 0.0015). The largest high- α gains come from the adaptive radius, which relaxes toward the nominal risk as α grows.

COVID-19 $10\times$ penalty matrix. In the high-stakes regime Λ_1 (Fig. 2e–h), RAC becomes brittle as α grows—realized loss inflates to 9.36 ± 0.15 at $\alpha = 0.10$ and critical mistakes reach $12.95 \pm 0.50\%$ (COVID-19 $\rightarrow$ No Action). The fixed loss-tail floor keeps DRO-ROCP essentially identical to ROCP, holding realized loss at 3.4196 ± 0.0004 and 0.0% critical mistakes across all hazardous labels, retaining heavy-tail safety without over-conservatism.

BDD100K driving. On the 8-state task (Fig. 3), DRO-ROCP delivers its largest gains at low α : at $\alpha = 0.001$ realized loss drops $4.906 \rightarrow 4.370$ and the risk certificate $5.664 \rightarrow 5.126$; it matches or beats ROCP at every α while tracking coverage (0.097 at $\alpha = 0.1$). At $\alpha = 0.05$ it cuts critical collision mistakes by $\sim 45\%$ vs. RAC (state 100: 9.6% vs. 14.8%).

4.3 ABLATION STUDIES

(1) Deconstructing $\eta(x, \alpha)$. We remove one radius component at a time on BDD100K (Fig. 4). Dropping the covariate term $u(x)$ raises low- α loss (4.83 vs. 4.37 at $\alpha = 0.001$), confirming instance-level entropy drives small- α precision. Removing the decay $m(\alpha)$ hurts high- α performance, and removing the loss-tail floor collapses the mean radius from 1.82 to 0.17 nats and regresses high- α loss ($+0.10$ at $\alpha = 0.10$). A constant scalar radius is inferior everywhere.

(2) Localized vs. global calibration. On BDD100K at $\alpha = 0.01$, Mondrian calibration ($K = 3$) collapses the per-bin miscoverage spread from 0.0249 to 0.0013 and lowers realized loss from 3.971 to 3.263 . Gating localization to $\alpha \leq 0.01$ avoids small-sample over-coverage at higher α (0.097 gated vs. 0.068 ungated at $\alpha = 0.10$).

(3) Out-of-set vs. in-set robustification. The 2×2 factorial in Table 2 shows the realized-loss gains on benign losses come entirely from out-of-set robustification (M_{DRO}); robustifying the in-set quantile (Q_t^{DRO}) inflates thresholds and raises loss (2.790 vs. 2.715). Hence $q_{\text{scale}} = 0$ (M_{DRO} only) is optimal for loss, while Q_t^{DRO} is an independent knob that further reduces critical mistakes (5.99%).

(4–6) Divergence geometry, estimators, hyperparameters. Comparing KL against total-variation (TV), χ^2 , and Wasserstein-1 (Appendix C, Fig. 5), KL uniquely guarantees heavy-tail safety: on COVID-19 $10\times$ at $\alpha = 0.05$, W_1 incurs 3.84% critical mistakes vs. 0.0% for KL and TV saturates on BDD100K, while χ^2 gives a $5.5\times$ speedup but weaker safety. A lightweight multi-temperature stack ($M = 5$) matches deep ensembles (2.728 vs. 2.745 at $\alpha = 0.05$), while over-dispersed MC-Dropout degrades quality (2.977). A 2D sweep over $\gamma, \kappa \in [0, 5]$ (Fig. 6) shows a broad flat optimum for $\gamma \geq 1, \kappa \geq 1$, validating the defaults $\gamma = 2, \kappa = 3$.

5 CONCLUSION

We presented DRO-ROCP, which bridges statistical decision theory, distributionally robust optimization, and split-conformal calibration. A dual-axis adaptive KL ambiguity set and a smooth 1D convex dual for the out-of-set risk $M_{\text{DRO}}(x, a)$ remove ROCP’s over-conservatism while guarding against base-model misspecification. Across medical diagnosis and driving-hazard benchmarks, DRO-ROCP reduces realized loss and critical mistakes and preserves exact finite-sample marginal coverage. **Limitations and future work.** DRO-ROCP assumes exchangeability for its coverage guarantee and adds modest overhead (~ 2 ms/query) for the 1D dual; extensions include group- and label-conditional ambiguity sets and localized DRO calibration under online distribution shift.

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A MATHEMATICAL PROOFS AND DUAL DERIVATIONS

A.1 DERIVATION OF THE 1D CONVEX DUAL FOR $M_{\text{DRO}}(x, a)$

We derive the Donsker–Varadhan dual for $M_{\text{DRO}}(x, a) = \sup_{q \in \mathcal{P}_\eta(x)} \mathbb{E}_{Y \sim q}[\ell(a, Y)]$, with $\mathcal{P}_\eta(x) = \{q \in \Delta_{\mathcal{Y}} : D_{\text{KL}}(q \parallel \bar{f}_x) \leq \eta\}$. The primal is

$$\max_{q \in \mathbb{R}_+^{|\mathcal{Y}|}} \sum_y q(y) \ell(a, y) \quad \text{s.t.} \quad \sum_y q(y) \ln \frac{q(y)}{\bar{f}_x(y)} \leq \eta, \quad \sum_y q(y) = 1. \quad (9)$$

With multipliers $\lambda \geq 0$ (KL constraint) and $\nu \in \mathbb{R}$ (normalization), the stationarity condition $\partial \mathcal{L} / \partial q(y) = \ell(a, y) - \lambda(\ln(q(y)/\bar{f}_x(y)) + 1) - \nu = 0$ gives $q^*(y) = \bar{f}_x(y) \exp((\ell(a, y) - \nu)/\lambda - 1)$. Enforcing normalization yields $\exp(\frac{\nu + \lambda}{\lambda}) = \sum_y \bar{f}_x(y) \exp(\ell(a, y)/\lambda)$, and substituting back gives $\mathbb{E}_{q^*}[\ell(a, Y)] - \lambda D_{\text{KL}}(q^* \parallel \bar{f}_x) = \lambda \ln \sum_y \bar{f}_x(y) \exp(\ell(a, y)/\lambda)$. Adding the penalty $\lambda \eta$ and minimizing over λ yields Eq. 6. The objective is a supremum of affine functions of $(\lambda, \lambda \eta)$, hence convex; strict convexity on $\lambda > 0$ follows from the positive-definite Hessian of the log-sum-exp term. ■

A.2 FINITE-SAMPLE MARGINAL COVERAGE

Proposition 1. *If $\{(X_i, Y_i)\}_{i=1}^n$ and $(X_{\text{test}}, Y_{\text{test}})$ are exchangeable and $\hat{\beta}_{1-\alpha}$ is computed by Eq. 8, then $\mathbb{P}(Y_{\text{test}} \in C_{\text{DRO}}(X_{\text{test}})) \geq 1 - \alpha$.*

Proof. Under exchangeability the dual scores $E_1, \dots, E_n, E_{\text{test}}$ are exchangeable. By the standard split-conformal argument (Angelopoulos & Bates, 2021), $\hat{\beta}_{1-\alpha} = \text{Quantile}(\{E_i\}, \frac{[(n+1)(1-\alpha)]}{n})$ implies $\mathbb{P}(E_{\text{test}} \leq \hat{\beta}_{1-\alpha}) \geq 1 - \alpha$. Since $E_{\text{test}} \leq \hat{\beta}_{1-\alpha} \iff Y_{\text{test}} \in C_{\text{DRO}}(X_{\text{test}})$ by definition of E_{test} , the claim follows. For gated localization the argument is applied within each entropy bin (a fixed measurable function of X), preserving exchangeability inside each group. □

B IMPLEMENTATION AND HYPERPARAMETERS

Table 3: Complete hyperparameter specification for DRO-ROCP.

Hyperparameter	Symbol	Value / Grid
Miscoverage level	α	$\{0.001, 0.005, 0.01, 0.02, 0.03, 0.05, 0.10\}$
Uncertainty multiplier	γ	2.0 (swept in $[0, 5]$)
Loss-tail floor scale	κ	3.0 (swept in $[0, 5]$)
Ambiguity cap	η_{cap}	0.6
Base radius floor	η_0	0.0 nats
Reference miscoverage	α_{ref}	0.02
Decay power	p	1.0
Multi-temperature stack	τ	$\{0.5, 0.7, 1.0, 1.4, 2.0\}$
Gated localized bins	K	3 entropy tertiles
Localization gate	–	$ \mathcal{Y} \geq 6$ and $\alpha \leq 0.01$
Dual search bounds	λ	$[10^{-4}, 100]$, golden-section search

C ADDITIONAL ABLATIONS

D EXTENDED TABULAR RESULTS

Tables 4–6 report all metrics across the full α grid (mean $\pm$ SE over 20 splits).

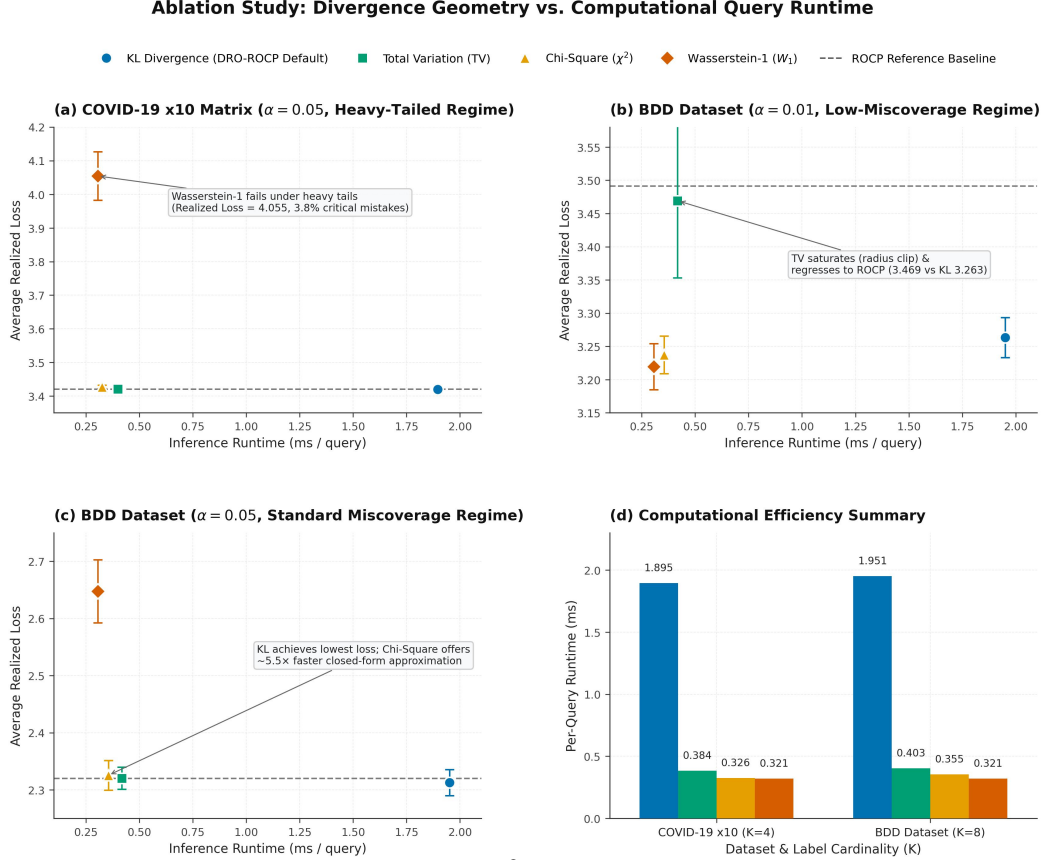


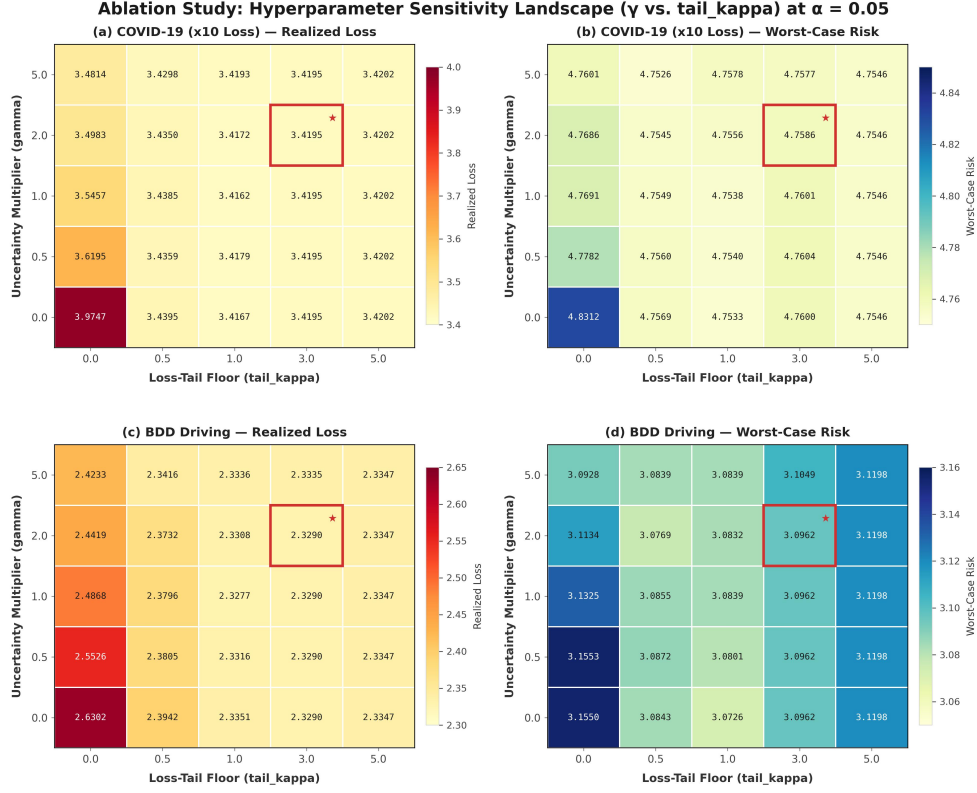
Figure 5: Divergence-geometry ablation (KL, TV, χ^2 , W_1). (a)–(c) realized loss vs. per-query runtime under heavy tails and varying α , anchored by the ROCP reference; (d) mean per-query runtime. KL uniquely delivers heavy-tail safety while maximizing decision performance; χ^2 is $\sim 5.5\times$ faster with competitive benign performance, while TV saturates and W_1 fails to protect against catastrophic losses. Bars are ± 1 SE over splits.

Table 4: Full results on COVID-19 under the base matrix Λ_0 (mean $\pm$ SE, 20 splits).

Method	$\alpha=.001$	$\alpha=.005$	$\alpha=.01$	$\alpha=.02$	$\alpha=.03$	$\alpha=.05$	$\alpha=.10$
<i>Realized loss ($\downarrow$)</i>							
DRO-ROCP	3.383 $\pm$.005	3.305 $\pm$.009	3.213 $\pm$.012	3.067 $\pm$.011	2.917 $\pm$.014	2.715 $\pm$.016	2.435 $\pm$.013
ROCP	3.386 $\pm$.005	3.312 $\pm$.009	3.227 $\pm$.012	3.086 $\pm$.012	2.944 $\pm$.016	2.750 $\pm$.017	2.484 $\pm$.016
RAC	3.379 $\pm$.005	3.295 $\pm$.008	3.192 $\pm$.010	3.026 $\pm$.011	2.874 $\pm$.013	2.667 $\pm$.014	2.364 $\pm$.010
<i>Worst-case risk certificate ($\downarrow$)</i>							
DRO-ROCP	5.866 $\pm$.012	5.671 $\pm$.015	5.461 $\pm$.017	5.106 $\pm$.017	4.756 $\pm$.025	4.281 $\pm$.037	3.422 $\pm$.027
ROCP	5.863 $\pm$.013	5.670 $\pm$.016	5.460 $\pm$.018	5.110 $\pm$.019	4.756 $\pm$.025	4.263 $\pm$.035	3.426 $\pm$.028
RAC	5.869 $\pm$.012	5.680 $\pm$.012	5.470 $\pm$.017	5.102 $\pm$.016	4.784 $\pm$.023	4.303 $\pm$.035	3.428 $\pm$.029

Table 5: Full results on COVID-19 under the $10\times$ penalty matrix Λ_1 (mean $\pm$ SE, 20 splits).

Method	$\alpha=.001$	$\alpha=.005$	$\alpha=.01$	$\alpha=.02$	$\alpha=.03$	$\alpha=.05$	$\alpha=.10$
<i>Realized loss ($\downarrow$)</i>							
DRO-ROCP	3.421 $\pm$.000	3.420 $\pm$.000	3.420 $\pm$.000	3.420 $\pm$.000	3.420 $\pm$.000	3.420 $\pm$.000	3.420 $\pm$.000
ROCP	3.421 $\pm$.000	3.421 $\pm$.000	3.421 $\pm$.000	3.421 $\pm$.000	3.420 $\pm$.000	3.421 $\pm$.000	3.422 $\pm$.000
RAC	3.439 $\pm$.007	3.588 $\pm$.022	3.816 $\pm$.030	4.393 $\pm$.046	4.952 $\pm$.075	6.113 $\pm$.132	9.357 $\pm$.149
<i>Worst-case risk certificate ($\downarrow$)</i>							
DRO-ROCP	5.874 $\pm$.012	5.695 $\pm$.014	5.510 $\pm$.021	5.226 $\pm$.012	5.020 $\pm$.014	4.754 $\pm$.012	4.334 $\pm$.011
ROCP	5.874 $\pm$.012	5.696 $\pm$.014	5.509 $\pm$.021	5.225 $\pm$.012	5.020 $\pm$.014	4.752 $\pm$.011	4.332 $\pm$.011
RAC	5.870 $\pm$.012	5.694 $\pm$.012	5.520 $\pm$.015	5.291 $\pm$.014	5.190 $\pm$.018	5.310 $\pm$.021	6.796 $\pm$.037



Key Observation: Performance remains exceptionally stable across $\text{tail_kappa} \geq 1.0$ and $\gamma \geq 1.0$. The default hyperparameter selection ($\gamma = 2.0$, $\text{tail_kappa} = 3.0$) resides in a flat, robust basin, confirming low sensitivity to precise tuning on heavy-tailed and complex decision tasks.

Figure 6: Hyperparameter landscape over γ (uncertainty multiplier) and κ (loss-tail floor) at $\alpha = 0.05$. (a, b) COVID-19 10x; (c, d) BDD100K. Red cells mark the default $(\gamma, \kappa) = (2, 3)$. Performance is stable across a broad basin ($\gamma \geq 1, \kappa \geq 1$), indicating no fragile tuning is required.

Table 6: Full results on BDD100K driving (mean $\pm$ SE, 20 splits).

Method	$\alpha=.001$	$\alpha=.005$	$\alpha=.01$	$\alpha=.02$	$\alpha=.03$	$\alpha=.05$	$\alpha=.10$
<i>Realized loss ($\downarrow$)</i>							
DRO-ROCP	4.370 $\pm$.141	4.003 $\pm$.145	3.263 $\pm$.030	2.528 $\pm$.088	2.320 $\pm$.019	2.313 $\pm$.023	2.315 $\pm$.023
ROCP	4.906 $\pm$.058	4.459 $\pm$.080	3.492 $\pm$.106	2.424 $\pm$.024	2.332 $\pm$.018	2.320 $\pm$.019	2.323 $\pm$.021
RAC	4.955 $\pm$.024	4.407 $\pm$.230	4.151 $\pm$.261	2.627 $\pm$.181	2.344 $\pm$.024	2.981 $\pm$.075	5.730 $\pm$.108
<i>Worst-case risk certificate ($\downarrow$)</i>							
DRO-ROCP	5.126 $\pm$.158	4.558 $\pm$.159	3.671 $\pm$.042	2.699 $\pm$.098	2.521 $\pm$.019	3.111 $\pm$.011	4.893 $\pm$.008
ROCP	5.664 $\pm$.078	4.933 $\pm$.078	3.837 $\pm$.110	2.615 $\pm$.045	2.553 $\pm$.021	3.136 $\pm$.014	4.887 $\pm$.010
RAC	5.728 $\pm$.039	4.974 $\pm$.317	4.683 $\pm$.337	2.901 $\pm$.215	2.441 $\pm$.014	3.126 $\pm$.019	5.383 $\pm$.009